

Samrat Talukdar

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SUMMARY

Backend Developer with 5+ years building production-ready automation bots, AI/ML systems, and Web3 fintech. Founded HoldPe and built Telegram ecosystems (2.5M+ users). Skilled in Python, backend APIs, computer vision, and blockchain integrations. Ships features quickly and improves system efficiency with measurable impact.

EDUCATION

Narula Institute of Technology, Agarpara | West Bengal Aug 2024 – May 2028
Bachelor of Technology in Computer Science and Engineering (AI & ML)

WORK EXPERIENCE

Founder | *HoldPe* | Remote Aug 2025 – Present

- Architected escrow microservices using NodeJS and Web3 integrations, projected to reduce manual reconciliation time by up to 80% in pilot workflows.
- Engineered transaction orchestration and monitoring pipelines, improving expected throughput 4x vs. manual processes (design & prototype stage).

AI Research Intern | *Numiq.AI* | Remote Aug 2025 – Present

- Fine-tuned LLMs and deployed streaming inference with optimized batching, cutting average request latency ~40% in internal benchmarks.
- Developed end-to-end voice pipeline (ASR → LLM → TTS), improving response relevance in A/B tests by ~25% (internal evaluation).

Software Developer | *Ultroid* | [GitHub](#) Jan 2022 – Oct 2024

- Optimized async Telethon modules and refactored I/O-heavy paths, increasing command throughput under concurrency by ~60% and lowering API error rate by ~30%.
- Added automated test suites and CI via GitHub Actions to reduce regressions and accelerate release cadence by ~50%.

Telegram Bot Developer | *Freelance* | Remote Apr 2020 – Oct 2024

- Designed and shipped 20+ production bots using Python, Telethon/Pyrogram, and MongoDB/Postgres, automating workflows for communities and commerce.
- Built Emilia, a modular group-management bot scaled to 2.5M+ users; streamlined moderation flows that reduced manual admin workload by ~80% and cut spam incidence by ~90% for target communities.
- Integrated Stripe & Braintree payment flows and crypto payment gateways to enable shop-bot checkout paths with a payment success rate improvement of ~35% after retry logic and validation.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C/C++, SQL, HTML5, CSS3

Frontend: ReactJS, NextJS, Tailwind CSS

Backend: Flask, FastAPI, NodeJS, REST APIs

Databases: MongoDB, PostgreSQL, MySQL, Redis

Tools/Cloud: Git, Docker, Apache, GitHub Actions, AWS

KEY PROJECTS

ProcurAI — Conversational Product Intelligence for Pumps & Motors August 2025

- Built end-to-end web scraping & ETL pipeline to ingest product data with ~90% extraction accuracy
- Converted structured JSON into embeddings stored in Pinecone for semantic search across 10k+ entries
- Implemented RAG-driven chatbot, improving answer relevance by ~45% vs keyword search
- Automated incremental updates & deduplication, reducing staleness by ~80% and cutting query latency by ~60%

EcoNova — AI-Powered Biodiversity & Threat Detection April 2025

- Implemented YOLOv8-based detection + LDA classification with ~90% accuracy on test datasets
- Deployed real-time alerts pipeline reducing hazard detection time by ~70% in prototype trials
- Added Generative AI reporting for automated incident summaries and rapid decision-making

ACHIEVEMENTS & LEADERSHIP

- **Research Publication:** "Leveraging Blockchain for Transparent Allocation of Government Funds" — Presented at Dr. Ram Manohar Lohiya National Law University, Lucknow (Feb 2025).
- **Stanford Certificate:** Machine Learning (Andrew Ng) | [Certificate](#)
- **Hackathons:** Certificates of Participation & Appreciation in multiple hackathons and seminars (AMUHACKS, Hack the Cosmos, Hack-o-NiT, IHMMC, TechGenia, etc.)